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§1 Tangency Condition for Orthogonal Axis Circles

Here we provide a complete geometric analysis and algebraic proof for the condition under which two circles centered on perpendicular coordinate axes intersect at exactly one point (i.e., touch each other).

Problem Statement

Prove that the circles

$$\mathcal{C}_1 : x^2 + y^2 + 2ax + c^2 = 0$$

and

$$\mathcal{C}_2 : x^2 + y^2 + 2by + c^2 = 0$$

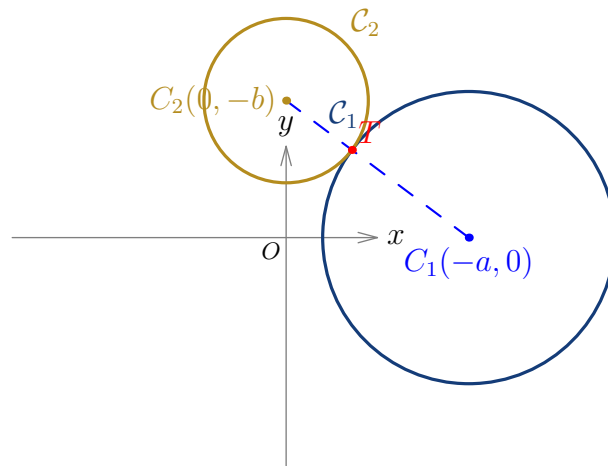
touch each other if and only if

$$\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}.$$

Source: Classical Coordinate Geometry

1.1 Geometric Visualization

Below is the geometric representation of the two circles $\mathcal{C}_1$ and $\mathcal{C}_2$ along with their respective centers C_1 and C_2 lying on the coordinate axes.



1.2 Rigorous Algebraic Proof

Proof. The general equation of a circle is expressed as:

$$x^2 + y^2 + 2gx + 2fy + k = 0$$

where the center is given by $C(-g, -f)$ and the radius is $r = \sqrt{g^2 + f^2 - k}$.

Step 1: Determine Centers and Radii For the first circle $\mathcal{C}_1 : x^2 + y^2 + 2ax + c^2 = 0$:

- Center $C_1 = (-a, 0)$

- Radius $r_1 = \sqrt{a^2 - c^2}$

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For instructional use only

For the second circle $\mathcal{C}_2 : x^2 + y^2 + 2by + c^2 = 0$:

- Center $C_2 = (0, -b)$

Squaring both sides eliminates the absolute value:

$$d^2 = (r_1 \pm r_2)^2 = r_1^2 + r_2^2 \pm 2r_1r_2$$

Substituting our expressions for d , r_1 , and r_2 :

$$a^2 + b^2 = (a^2 - c^2) + (b^2 - c^2) \pm 2\sqrt{a^2 - c^2}\sqrt{b^2 - c^2}$$

Simplifying by subtracting $a^2 + b^2$ from both sides results in:

$$0 = -2c^2 \pm 2\sqrt{a^2 - c^2}\sqrt{b^2 - c^2}$$

$$2c^2 = \pm 2\sqrt{a^2 - c^2}\sqrt{b^2 - c^2}$$

Dividing both sides by 2:

$$c^2 = \pm\sqrt{a^2 - c^2}\sqrt{b^2 - c^2}$$

Step 4: Algebraic Reduction Squaring both sides once more gives:

$$(c^2)^2 = (a^2 - c^2)(b^2 - c^2)$$

$$c^4 = a^2b^2 - a^2c^2 - b^2c^2 + c^4$$

Eliminating c^4 from both sides:

$$0 = a^2b^2 - a^2c^2 - b^2c^2$$

$$a^2c^2 + b^2c^2 = a^2b^2$$

Step 5: Final Inversion Form Dividing the entire equation by the non-zero product $a^2b^2c^2$:

$$\frac{a^2c^2}{a^2b^2c^2} + \frac{b^2c^2}{a^2b^2c^2} = \frac{a^2b^2}{a^2b^2c^2}$$

$$\frac{1}{b^2} + \frac{1}{a^2} = \frac{1}{c^2}$$

Rearranging the terms yields the desired identity:

$$\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$$

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